

Appendix: Rigorous Derivations of Fundamental Constants

The following section provides the fully rigorous proofs for the emergence of constants and key structures, built from the SAT Minimal Ontology (see conversation history).

Appendix A: Derivation of the Speed of Light c

Statement

We prove that the SAT framework, starting only from its primitive fields u^μ and θ_4 , necessarily yields a maximal causal speed scale c arising naturally from the foliation structure and strain tensor, without external input.

The Speed of Light:

In SAT, c emerges as the maximal causal speed from first principles, without external input.

Appendix B: Derivation of Planck's Constant $\hbar$

Statement

We prove that the SAT framework, based only on the primitive field $\psi(x)$ and the compactness of its domain, necessarily implies the existence of a fundamental action quantum $\hbar$, without external input or tuning.

Planck's Constant:

In SAT, $\hbar$ emerges necessarily as a quantum of action from the compactness of ψ , without external input.

Appendix C: Derivation of the Elementary Charge e

Statement

We prove that the SAT framework, using only the primitive field $\psi(x)$ and its minimal coupling to the emergent gauge field $A_\mu(x)$ under local $U(1)$ gauge symmetry, necessarily implies the quantization of electric charge in integer multiples of a fundamental unit e , without external input or tuning.

The Elementary Charge:

In SAT, e emerges necessarily as a quantized coupling constant from the ψ - $U(1)$ structure, without external input.

Appendix D: Derivation of the Fine-Structure Constant α **Statement**

We prove that the SAT framework, using only its internally derived quantities $\hbar$, e , and the emergent speed of causal propagation c , necessarily constructs the dimensionless fine-structure constant α without external input or tuning.

The Fine-Structure Constant:

In SAT, the fine-structure constant α emerges necessarily as a dimensionless ratio of internally derived quantities e , $\hbar$, and c , without external input or tuning.

Appendix E: Derivation of Newton's Gravitational Constant G **Statement**

We prove that the SAT framework, based only on the strain dynamics of the preferred time-flow vector u^μ and the misalignment angle θ_4 , necessarily gives rise to an emergent gravitational coupling constant G in the low-energy limit, without external input or tuning.

Newton's Gravitational Constant:

In SAT, G emerges necessarily from the dynamics of the foliation strain tensor $S_{\mu\nu}$ and vacuum structure, without external input or tuning.

Appendix F: Derivation of the Electron Mass m_e **Statement**

We prove that the SAT framework, based on the internal clock phase field $\psi(x)$ and the topological twist field $\tau(x)$, naturally produces a discrete mass spectrum for fermions, with the electron mass m_e emerging as the minimal excitation without external input or arbitrary parameters. The mass spectrum is governed by the topological invariant $\mathbf{Q} = \mathbf{L}_{\text{wind}} + \mathbf{L}_{\text{link}} + \mathbf{W}_{\text{writhe}}$.

The Electron Mass:

In SAT, the electron mass m_e emerges as the minimal stable topological excitation, determined by the winding structure of $\psi(x)$ and the suppression factor $m_{\text{eff}} = m_0/Q$, without external input or arbitrary parameters.